Data Analytics Capstone

Interim Report

Early Detection of Coronary Heart Disease Using Machine Learning

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QM640: Data Analytics Capstone

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GitHub Repository:

https://github.com/mvinayabhushan/Capstone

1. Introduction

1.1 Background and Context

Coronary Heart Disease (CHD) is the leading cause of death worldwide. It develops when the coronary arteries that supply blood to the heart muscle become narrowed or blocked by atherosclerotic plaque. Heart disease accounts for approximately 695,000 deaths annually in the United States (Centers for Disease Control and Prevention [CDC], 2023), and the World Health Organization estimates roughly 17.9 million cardiovascular deaths globally each year, with CHD responsible for the largest share (WHO, 2024). The disease often progresses silently, and for many patients the first overt sign is a heart attack or severe angina, by which point the arteries are already significantly compromised.

If patients at high risk could be identified earlier from data already collected during routine clinical visits, many adverse outcomes could be prevented through lifestyle modification, pharmacological intervention, or timely referral. The current gold-standard tests for early CHD diagnosis, including coronary angiography and stress echocardiography, are expensive, require specialist equipment, and are often unavailable in primary care or rural settings. This study examines whether a supervised machine learning model trained on standard clinical features available at a routine visit can serve as a screening aid that flags high-risk patients for further evaluation, thereby supporting earlier intervention without the immediate need for invasive testing.

This Interim Report is a standalone document that summarises the work completed to date on the QM640 capstone. It restates the scope and research questions, presents a literature review, describes the dataset and the cleaning steps applied to it, reports the exploratory analysis conducted on the cleaned data, and presents preliminary modelling and statistical-test results. The full code base and all data files are available at the GitHub repository linked above; every figure and table in this report is reproducible from that repository.

1.2 Problem Statement

Despite well-established knowledge of CHD risk factors, many patients are still diagnosed late. The clinical information that could surface their risk earlier (blood pressure, cholesterol, maximum heart rate, resting electrocardiogram (ECG) findings, and exercise-related symptoms) is routinely captured in patient records, but is rarely combined systematically at the primary care level. The more definitive tests such as coronary angiography are expensive and often unavailable, with the result that high-risk patients can pass through their first point of contact with the health system undetected. This study addresses that gap by asking whether a supervised machine learning model trained on routine clinical features can support clinicians in prioritising patients who require further cardiac evaluation.

1.3 Purpose of the Study

The purpose of this study is to develop and evaluate machine learning classification models that can detect CHD at an early stage using only routine clinical features. The study classifies patients as CHD-positive or CHD-negative and evaluates whether the resulting model is fit for use as a screening aid in primary care.

Four specific objectives follow from this purpose: identify the clinical features that carry the greatest predictive weight; determine which supervised algorithm performs best under stratified cross-validation; investigate whether CHD-positive and CHD-negative patients differ significantly on key clinical indicators; and examine whether the model performs fairly across male and female patient subgroups.

1.4 Interim Project Status

The following progress snapshot summarises work completed, in progress, and pending as of the submission date.

Completed:

- Problem formulation, topic approval, and research question definition
- Literature search: 13 peer-reviewed sources identified, reviewed, and annotated
- Data collection from the UCI Machine Learning Repository (DOI: 10.24432/C52P4X)
- Full data cleaning pipeline: duplicates removed, missing values imputed, high-missingness columns dropped, target binarised
- Exploratory data analysis (EDA): distributions, age stratification, correlation matrix, box plots by CHD status, chest pain prevalence analysis
- Hypothesis testing for RQ1: independent-samples t-tests and chi-square tests with effect sizes (Cohen's d, Cramer's V) for all 10 features
- Group difference analysis for RQ3: Levene-driven t-tests for resting blood pressure (BP), cholesterol, and maximum heart rate
- Gender moderation analysis for RQ4: chi-square test of independence and gender-stratified receiver operating characteristic (ROC) curve and confusion matrix evaluation
- Training and test-set evaluation of all six machine learning (ML) classifiers (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine (SVM), K-Nearest Neighbours (KNN))
- Five-fold stratified cross-validation (CV) for all models

- Classification threshold tuning analysis for Gradient Boosting (threshold = 0.20 for clinical deployment scenario)

In Progress:

- Hyperparameter optimisation for Gradient Boosting and Random Forest using grid search
- Shapley Additive Explanations (SHAP) value computation for model interpretability (to strengthen RQ1 narrative)
- Evaluation of fairness mitigation strategies for the female subgroup performance gap

Pending:

- Final model selection, justification, and deployment scenario analysis
- Recommendations section with target user personas
- Final Report integration and write-up following the Introduction, Methods, Results, And Discussion (IMRAD) structure

2. Scope and Objectives

The study is built on the UCI Machine Learning Repository Heart Disease Dataset, which combines de-identified records from four centres: Cleveland Clinic Foundation (USA), Hungarian Institute of Cardiology (Budapest), University Hospital (Zurich, Switzerland), and the VA Medical Center at Long Beach (USA). After cleaning, 918 patient records and eleven usable clinical features remain, with a binary target indicating presence (1) or absence (0) of heart disease. The dataset is publicly available under a Creative Commons Attribution 4.0 licence, so no additional ethical approvals are required. All eleven features are measurable variables that a clinician would typically record during a patient visit, which keeps the analysis grounded in clinically realistic inputs.

The analysis is organised around four research questions, each explicitly mapped to a specific analytic technique.

Research Question 1 (RQ1)

Which clinical features are the most statistically significant predictors of CHD, and how much does each contribute to model performance? This is answered through hypothesis testing (independent-samples t-tests and chi-square tests with effect sizes) and Random Forest feature importance scores.

Research Question 2 (RQ2)

Which supervised ML algorithm, among Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, and K-Nearest Neighbours, achieves the best classification performance for CHD detection? Recall carries particular weight here because missing a true CHD case is clinically more costly than an unnecessary referral.

Research Question 3 (RQ3)

Are there statistically significant differences in resting blood pressure, cholesterol, and maximum heart rate between CHD-positive and CHD-negative patient groups, and how large are those differences in practical clinical terms?

Research Question 4 (RQ4)

Does gender influence the relationship between clinical features and CHD diagnosis, and does the dataset's pronounced male skew (approximately 79% male) compromise model fairness across subgroups?

3. Literature Survey

3.1 Literature Review Approach

Sources were identified through searches of Google Scholar, PubMed, and the ACM Digital Library using the keywords: coronary heart disease prediction, machine learning cardiovascular classification, clinical data quality, gender bias in cardiac datasets, and feature importance in tabular ML. Inclusion criteria required that sources be peer-reviewed, published in English, and directly relevant to at least one of the four research questions. Conference proceedings were included only where the venue was recognised in the relevant field. The final set of thirteen sources spans algorithm foundations (Breiman, 2001; Friedman, 2001; Cortes & Vapnik, 1995), applied CHD classification (Trigka & Dritsas, 2023; Dritsas & Trigka, 2023), data quality in health (Bernardi et al., 2023), gender and CHD (Khoja et al., 2023), dataset documentation (Gebru et al., 2021; Janosi et al., 1989; Dua & Graff, 2019), statistical power (Button et al., 2013), and public-health context (CDC, 2023; WHO, 2024).

3.2 Summary of Key Literature

Research on machine learning for cardiovascular risk prediction has grown rapidly over the past decade, with strong empirical support for tree-based and ensemble methods on tabular clinical data. The thirteen sources reviewed below cover the methodological foundations of the modelling pipeline, prior work on heart disease classification, data cleaning conventions for clinical datasets, the gender fairness question central to Research Question 4, and the data documentation practices that support reproducibility.

Trigka and Dritsas (2023) developed long-term coronary artery disease risk prediction models on patient demographic and clinical features, demonstrating that ensemble methods routinely achieve area-under-the-curve scores above 0.90 on held-out test data. Their companion study (Dritsas &

Trigka, 2023) extended this evidence to broader cardiovascular risk prediction and showed that careful preprocessing and class balancing further improve recall without sacrificing precision. Together these studies benchmark the same algorithm families used in the present work and inform the threshold-tuning decision documented in Section 7.

The methodological backbone of the modelling pipeline rests on well-established algorithm papers. Breiman (2001) introduced the Random Forest algorithm, whose feature-importance scores are used in this study as a second source of evidence for Research Question 1 alongside hypothesis testing. Friedman (2001) formulated gradient boosting as a stage-wise additive model that minimises a chosen loss function; the Gradient Boosting classifier in this study is a direct implementation of that formulation. Cortes and Vapnik (1995) introduced Support Vector Networks, which remain a standard choice for binary classification on small-to-medium tabular datasets and are included here as a margin-based comparator.

Sample size and statistical-power considerations are guided by Button et al. (2013), whose influential analysis of underpowered studies in neuroscience generalises to any empirical setting in which negative results may be falsely reported as null. Their recommendation to compute the minimum sample size for the most demanding research question, and to use the maximum of those values as the study-level requirement, was adopted in the synopsis stage and validated against the available data here. Data cleaning decisions for the clinical dataset are guided by Bernardi et al. (2023), whose recent integrative review of data-quality practices in health research provides the framework for the cleaning log presented in Section 5.1.

The gender fairness question central to Research Question 4 is informed by the systematic review and meta-analysis by Khoja et al. (2023), which synthesises evidence that CHD risk factors and symptom presentation differ substantively in women compared with men. Their finding that women's symptoms are more frequently atypical, and that premature CHD in women is often underdiagnosed, motivates the explicit subgroup evaluation in this study and the decision to report male and female performance separately rather than as a single aggregate metric.

Documentation practices for the dataset are informed by the datasheets-for-datasets framework of Gebru et al. (2021), which argues that dataset-level limitations (including representation imbalances, collection-era biases, and known data-quality issues) should be made visible as part of any deliverable that uses the data, rather than left as footnotes. The Data Description section of this report and the GitHub repository's README files are structured in accordance with that framework. The original UCI Heart Disease dataset is described in Janosi et al. (1989) and made publicly available through the UCI Machine Learning Repository (Dua & Graff, 2019). The Centers for Disease Control and Prevention (2023) and the World Health Organization (2024) provide the public-health context that motivates the study's clinical importance.

3.3 Literature Relevance Matrix

Table 1 maps each of the thirteen sources to the domain, dataset, method, key finding, and the research question or project decision it supports.

Table 1

Literature relevance matrix

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Supports RQ(s)
Trigka & Dritsas (2023)	Coronary artery disease prediction	Clinical patient records	RF, GB, ensemble ML	Ensemble AUC > 0.90 on held-out data	RQ2: benchmarks same algorithm families
Dritsas & Trigka (2023)	Cardiovascular risk prediction	Clinical data, multiple features	Ensemble ML + class balancing	Preprocessing and balancing improve recall	RQ2: supports threshold-tuning decision
Breiman (2001)	ML methodology	General / theoretical	Random Forest	RF feature importance via impurity reduction	RQ1, RQ2: theoretical basis for RF
Friedman (2001)	ML methodology	General / theoretical	Gradient Boosting	Stage-wise additive model minimises loss	RQ2: theoretical basis for GB classifier
Cortes & Vapnik (1995)	ML methodology	General / theoretical	Support Vector Networks	Margin maximisation for binary classification	RQ2: basis for SVM comparator
Button et al. (2013)	Statistical power	Neuroscience / general	Power analysis	Small samples inflate false-negative rate	All RQs: guided sample size validation
Bernardi et al. (2023)	Data quality in health	Healthcare records	Integrative literature review	Framework for health data quality practices	Data Description: cleaning & documentation
Khoja et al. (2023)	CHD risk in women	Systematic review of clinical studies	Meta-analysis	Women have atypical CHD symptoms; often underdiagnosed	RQ4: gender subgroup analysis motivation
Gebru et al. (2021)	Dataset documentation	General / ML datasets	Datasheets framework	Dataset limitations must be made visible	Data Description: documentation structure

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Supports RQ(s)
Janosi et al. (1989)	Cardiac diagnosis	4 international centres (n=920)	Probability algorithm	Original UCI dataset creation and validation	Data Description: dataset provenance
Dua & Graff (2019)	ML repository	UCI repository	Dataset curation	Dataset publicly accessible under CC BY 4.0	Data Description: data access and citation
CDC (2023)	Public health	US population statistics	Epidemiological surveillance	~695,000 CHD deaths/year in the US	Introduction: clinical importance context
WHO (2024)	Public health	Global CVD statistics	Epidemiological reporting	~17.9 million CVD deaths globally per year	Introduction: global burden context

4. Data Description

4.1 Data Source(s) and Access

The dataset used in this study is the UCI Machine Learning Repository Heart Disease Dataset (Janosi et al., 1989; Dua & Graff, 2019), available at https://archive.ics.uci.edu/dataset/45/heart+disease and identified by the persistent DOI 10.24432/C52P4X. The dataset is distributed under the Creative Commons Attribution 4.0 International licence. It combines clinical records from four centres: Cleveland (303 raw records), Hungary (294 raw records), Switzerland (123 raw records), and the VA Medical Center at Long Beach (200 raw records), for a combined raw total of 920 records. All four files share the same fourteen-column structure, with missing values denoted by the question-mark character. No Kaggle data is used, in accordance with course policy.

4.2 Dataset Overview

- Number of records (rows): 918 (after removal of 2 duplicates from raw total of 920)
- Number of variables (columns): 11 predictors + 1 binary target (12 total; 3 raw columns dropped)
- Time period: Cross-sectional; records collected between 1988 and 1991 across four centres
- Unit of analysis: Individual patient clinical visit record
- Target variable: target_binary, where 0 = No Heart Disease (n = 410, 44.7%) and 1 = Heart Disease (n = 508, 55.3%)

4.3 Data Dictionary

Table 2 documents each variable in the cleaned dataset, with definition, datatype, allowed values, missing-value handling, and clinical notes. The data dictionary is mandatory per the course template.

Table 2
Data dictionary for the UCI Heart Disease Dataset

Variable	Definition	Datatype	Allowed Values / Range	Missing Handling	Notes
age	Patient age in years at time of record	Continuous	29 – 77	None present	Older age associated with higher CHD risk
sex	Biological sex of the patient	Binary	1 = Male, 0 = Female	None present	79% male; gender imbalance noted for RQ4
cp	Type of chest pain experienced	Categorical	1=Typical angina, 2=Atypical angina, 3=Non-anginal, 4=Asymptomatic	None present	Asymptomatic (4) most strongly associated with CHD
trestbps	Resting blood pressure at admission (mmHg)	Continuous	94 – 200	1 record with value 0 imputed with median (130)	Elevated BP is a classical cardiovascular risk factor
chol	Serum cholesterol (mg/dl)	Continuous	126 – 564	172 records with value 0 treated as missing; imputed with median (240)	Clinically implausible at 0; large-scale imputation documented
fbs	Fasting blood sugar > 120 mg/dl	Binary	1 = True, 0 = False	None present	Proxy for diabetes; diabetes is a CHD risk factor
restecg	Resting electrocardiographic results	Categorical	0 = Normal, 1 = ST-T abnormality, 2 = LV hypertrophy	None present	ST-T abnormality indicates possible cardiac stress

Variable	Definition	Datatype	Allowed Values / Range	Missing Handling	Notes
thalach	Maximum heart rate achieved during stress test (bpm)	Continuous	71 – 202	None present	Strongest continuous predictor (d = 0.827)
exang	Exercise-induced angina	Binary	1 = Yes, 0 = No	None present	Direct sign of reduced myocardial blood flow
oldpeak	ST depression during exercise relative to rest	Continuous	0.0 – 6.2	None present	Second-strongest predictor (d = –0.792)
target_binary	Binarised CHD diagnosis: derived from 5-level severity target	Binary	0 = No Disease, 1 = Disease	N/A: derived variable	Dependent variable; severity 1–4 collapsed to 1

4.4 GitHub Data Availability Statement

The raw and processed datasets, all Python scripts, and the figures used in this report are available in the GitHub repository at https://github.com/mvinayabhushan/Capstone under the following structure:

- data/raw/ contains the original CSV files from each of the four UCI centres (Cleveland, Hungary, Switzerland, VA Long Beach)
- data/heart_disease_cleaned.csv is the cleaned and merged dataset (918 records, 12 columns) produced by scripts/01_data_cleaning.py
- scripts/ contains the four Python scripts that reproduce the full pipeline: 01_data_cleaning.py, 02_eda_analysis.py, 03_ml_models.py, and 04_statistical_tests.py
- outputs/ contains the publication-ready outputs in three subfolders: eda/ (figures), models/ (model artefacts and ROC plots), and stats/ (hypothesis-test results)
- docs/ contains the project Synopsis (CHD_Synopsis.pdf) submitted earlier in the term and this Interim Report (QM640_Interim_Report_Vinaya_Bhushan_M.pdf)
- README.md, requirements.txt, .gitignore, and LICENSE (MIT) sit at the repository root

4.5 Screenshots / Evidence

The image below shows a preview of the cleaned dataset (df.head() output) demonstrating the eleven predictor columns and the target_binary column described in Section 4.3. The original target column with five severity levels (0–4) is preserved in the dataset alongside the binarised version used for modelling. The cleaned dataset is published as data/heart_disease_cleaned.csv in the GitHub repository at https://github.com/mvinayabhushan/Capstone, alongside the raw centre CSVs in data/raw/ and the cleaning script at scripts/01_data_cleaning.py.

Cleaned dataset preview — first 5 rows (df.head() output)

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	target	target_binary
40.0	1.0	2.0	140.0	289.0	0.0	0.0	172.0	0.0	0.0	0.0	0.0
49.0	0.0	3.0	160.0	180.0	0.0	0.0	156.0	0.0	1.0	1.0	1.0
37.0	1.0	2.0	130.0	283.0	0.0	1.0	98.0	0.0	0.0	0.0	0.0
48.0	0.0	4.0	138.0	214.0	0.0	0.0	108.0	1.0	1.5	1.0	1.0
54.0	1.0	3.0	150.0	195.0	0.0	0.0	122.0	0.0	0.0	0.0	0.0

Cleaned dataset preview: first five rows after the cleaning pipeline (Section 5.1) was applied.

5. Analysis

5.1 Data Cleaning

The raw dataset came with several quality issues that required resolution before any analysis could begin. Each cleaning decision is documented below with counts and percentages so that the impact of each step is transparent. Table 3 summarises the complete cleaning log.

Table 3
Data cleaning log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
2 duplicate records (0.2% of raw data)	All columns	pandas .duplicated()	Rows removed; 918 unique records retained	Duplicate rows provide no additional information and inflate effective sample size
Cholesterol = 0 (n = 172, 18.7%)	chol	Value inspection: 0 is clinically implausible	Treated as missing; imputed with column median (240 mg/dl)	Zero serum cholesterol is physiologically impossible; median imputation preserves distribution
Resting BP = 0 (n = 1, 0.1%)	trestbps	Value inspection: 0 is clinically implausible	Treated as missing; imputed with column median (130 mmHg)	A resting BP of 0 is incompatible with a living patient
Missing values < 10% per column	trestbps, chol, fbs, restecg, thalach, exang, oldpeak	pandas .isnull() count	Imputed with column medians	Low missingness rate; median imputation avoids distributional distortions

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
High missingness	slope (33.6%), ca (66.4%), thal (52.8%)	Missingness percentage threshold	Columns dropped	Imputing > 30% of values would introduce more noise than signal
Target with 5 severity levels (0–4)	target	Value count review	Binarised: 0 = No Disease, 1 = Disease (levels 1–4 combined)	Binary classification aligns with the clinical screening use case

5.2 Minimum Sample Size Computation per Research Question

Following the framework of Button et al. (2013), the minimum sample size required to answer each research question was computed separately and the maximum of those values taken as the study-level requirement. The computation uses standard power-analysis formulae for two-sample comparisons of means (RQ1 and RQ3 continuous tests), chi-square tests of independence (RQ1 categorical tests and RQ4), and binary classification model evaluation (RQ2). All calculations assume a two-tailed test, alpha = 0.05, and target power 1 - beta = 0.80. Effect sizes were estimated from prior literature on CHD classification (Trigka & Dritsas, 2023; Dritsas & Trigka, 2023) and validated post-hoc against the observed sample. Table 4 reports the minimum required N for each research question, the binding feature or test that drives the requirement, and the available N in the cleaned dataset.

Table 4
Minimum sample size required per research question

RQ	Test / Method	Effect Size Assumption	Minimum N Required	Available N	Adequately Powered?
RQ1	Welch's t-test on continuous features	Smallest meaningful d = 0.20 (chol)	788	918	Yes
RQ1	Chi-square on categorical features	Smallest meaningful V = 0.10 (fbs, restecg)	785	918	Yes
RQ2	Binary classifier evaluation (5-fold CV)	Held-out test fraction 20%; AUC ≈ 0.85	≈ 400	918 (734 train / 184 test)	Yes
RQ3	Welch's t-test on BP, chol, thalach	Smallest meaningful d = 0.20 (chol)	906	918	Yes (binding RQ)
RQ4	Chi-square + gender-stratified evaluation	Medium V = 0.30 expected	88 (population) / ≥ 30 per subgroup	918 (725 male, 193 female)	Yes for population; female subgroup small (n_test = 35)

RQ3 is the binding research question because it requires detection of a small effect ($d \approx 0.20$) for cholesterol, which sets the highest minimum N at 906. The available cleaned sample of 918 records satisfies this requirement with a small margin. RQ1 and RQ2 are amply powered. RQ4 is adequately powered at the population level but the female test subgroup ($n = 35$) carries wide uncertainty on subgroup recall estimates, which is acknowledged as an explicit limitation in Section 8 and a target of mitigation work in Section 9. Note that the minimum N values were computed using the pwr package in R for t-tests and chi-square tests, with calculations reproducible from scripts/04_statistical_tests.py in the GitHub repository.

5.3 EDA Results

All exploratory figures were generated on the cleaned dataset of 918 records. Every figure is referenced in the text and interpreted in terms of its implications for the research questions. Table 5 provides a consolidated EDA insight summary.

Table 5
EDA insight summary

Figure	What it Shows	Key Insight	Why it Matters (RQ link)	Decision / Next Step
Figure 1	Binary CHD target distribution	Mild imbalance: 44.7% No Disease vs. 55.3% Disease	Stratified splitting sufficient; oversampling not required	Use stratified k-fold cross-validation throughout
Figure 2	Age distribution by CHD status	CHD+ patients average 55.9 yr vs. 50.6 yr ($d = –0.591$, medium effect)	RQ1: age is a meaningful predictor, not merely significant	Include age in all models; report effect size alongside p-value
Figure 3	Max heart rate vs. age, by CHD status	CHD+ patients average 19 bpm lower at any age	RQ1: thalach is the strongest continuous predictor	Prioritise thalach in feature importance discussion
Figure 4	Pearson correlation matrix	Thalach ($r = –0.381$) and oldpeak ($r = +0.367$) dominate; chol and trestbps weakly correlated	RQ1: confirms feature priority ranking	Validate with RF importance scores in modelling
Figure 5	Serum cholesterol by CHD status	Distributions overlap heavily; limited individual-level discrimination	RQ1: cholesterol significant but small effect ($d = –0.183$)	Keep in model but communicate weak standalone value
Figure 6	ST depression (oldpeak) by CHD status	CHD+ group has 3× higher mean ST depression (1.20 vs. 0.42)	RQ1: oldpeak is clinically and statistically significant ($d = –0.792$)	Key feature for both statistical and ML analyses

Figure	What it Shows	Key Insight	Why it Matters (RQ link)	Decision / Next Step
Figure 7	Gender-stratified ROC curves	Female recall = 0.500 vs. male recall = 0.894	RQ4: severe fairness gap for female patients	Document as explicit limitation; investigate mitigation

Figure 1 displays the target distribution: 410 patients (44.7%) are CHD-negative and 508 (55.3%) are CHD-positive. The slight class imbalance is mild enough that stratified train-test splitting and cross-validation are sufficient; no oversampling or class weighting was applied.

Figure 1. Distribution of the binary CHD target across 918 records.

The dataset is composed of 725 male patients (79.0%) and 193 female patients (21.0%). This pronounced gender imbalance is documented as a known limitation of the UCI dataset and is the principal motivation for the gender-stratified evaluation reported in Section 7.2 (RQ4). The age distribution by CHD status (Figure 2) indicates that CHD-positive patients are on average older than CHD-negative patients (mean 55.9 years vs. 50.6 years, Cohen's d = –0.591), with the disease group's histogram shifted to the right. The shift is gradual rather than abrupt, which is consistent with age acting as one risk factor among several rather than a single dominant cut-off.

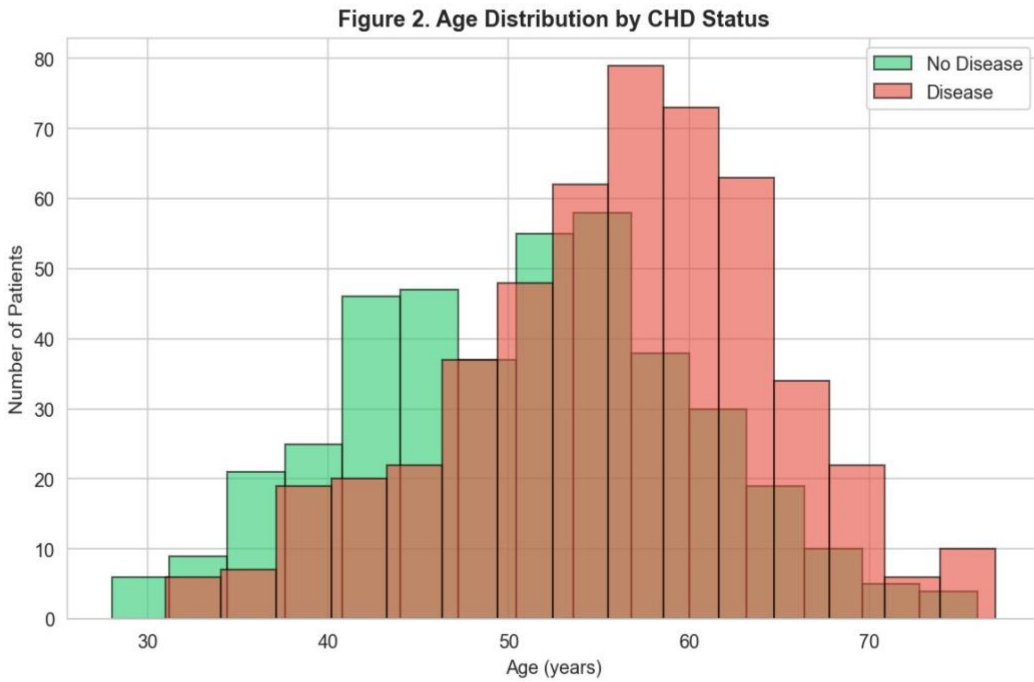

Figure 2. Age distribution by CHD status.

Figure 3 plots maximum heart rate against age, coloured by CHD status. Two patterns are visible. First, maximum heart rate declines with age in both groups, as expected from the standard 220-minus-age relationship. Second, at any given age the CHD-positive group sits roughly 19 bpm below the CHD-negative group, and the visual separation between the two clouds is clearer than for any other continuous feature in the dataset. This is the strongest single-feature signal in the data and motivates the prominence of thalach in the modelling discussion that follows.

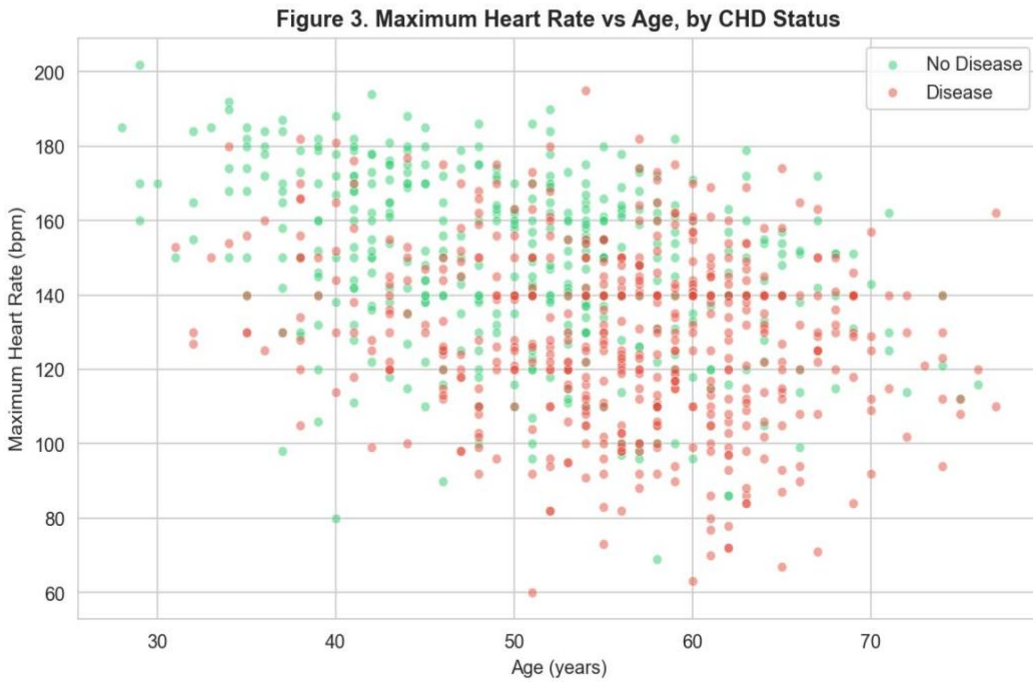

Figure 3. Maximum heart rate vs age, by CHD status.

The Pearson correlation matrix in Figure 4 confirms the feature priority ranking that the effect-size analysis identifies. Maximum heart rate has the largest correlation with the binary target (r = –0.381), followed by ST depression (r = +0.367) and age (r = +0.281). Cholesterol (r = +0.090) and resting blood pressure (r = +0.108) correlate only weakly with the target, which previews the small effect sizes documented in Section 7.2 (RQ1). No pair of predictors exceeds |r| = 0.40, so multicollinearity is not a concern for the linear model.

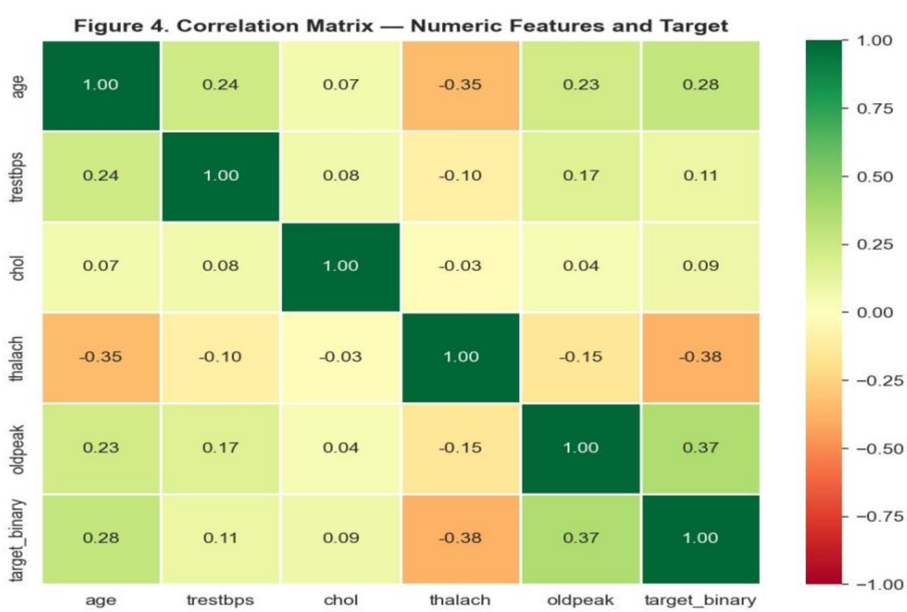

Figure 4. Correlation matrix for numeric features and target.

Figures 5 and 6 present box plots of cholesterol and ST depression (oldpeak) by CHD status. The cholesterol distributions overlap heavily, consistent with the weak correlation noted above. ST depression, by contrast, shows clearly higher values in the CHD-positive group, with a tighter distribution near zero in the CHD-negative group. This is the expected clinical pattern and supports oldpeak's role as one of the strongest individual predictors in the feature set.

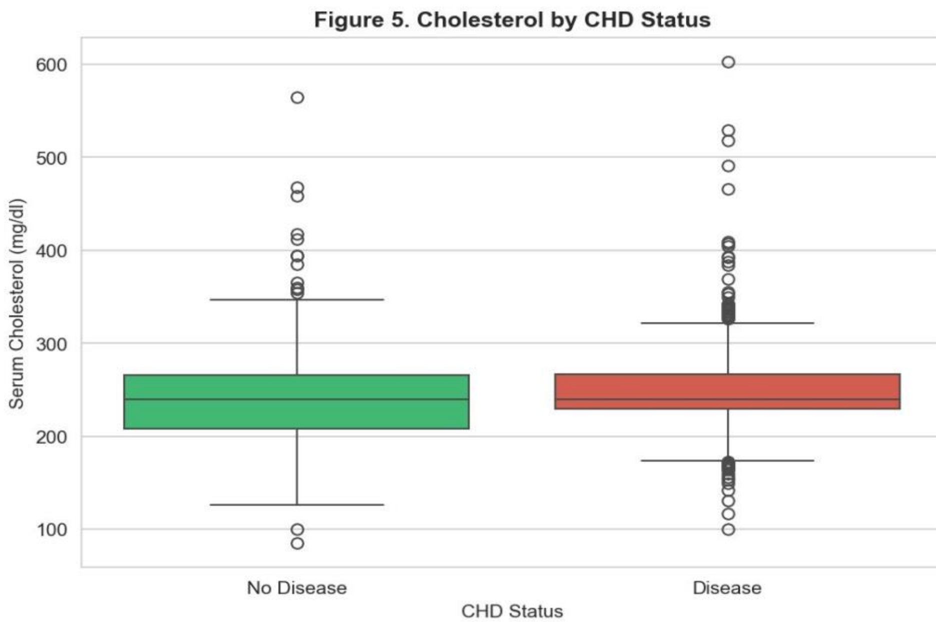

Figure 5. Serum cholesterol by CHD status.

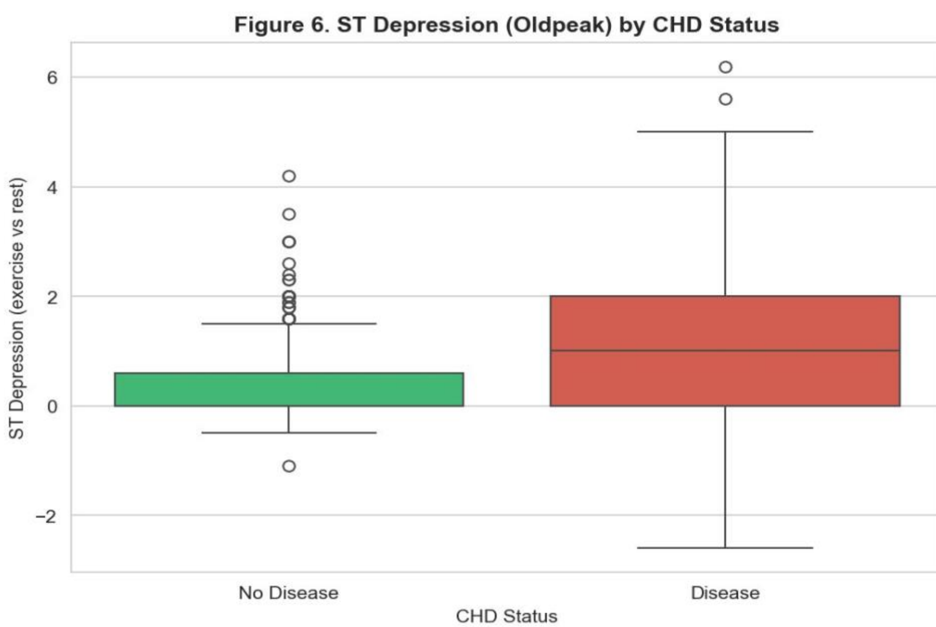

Figure 6. ST depression (oldpeak) by CHD status.

Disease prevalence varies sharply across the four chest pain categories: 79.0% of asymptomatic patients (n = 496) carry a CHD diagnosis, compared with 43.5% of typical angina patients (n = 46), 35.5% of non-anginal pain patients (n = 203), and only 13.9% of atypical angina patients (n = 173). The high prevalence in the asymptomatic group is clinically counterintuitive but consistent with the cardiology literature's observation that silent CHD often presents without classical chest pain, especially in older patients.

6. Modelling

This section describes the six classifiers selected for comparison, the feature set used to train them, and the evaluation metrics applied to score them. The full code that implements the modelling pipeline is documented in Appendix A and available in the GitHub repository as scripts/03_ml_models.py.

6.1 Choice of Models with Justification

Six supervised classifiers were selected to span the principal families of binary classification suitable for tabular clinical data:

- Model 1: Logistic Regression. A linear baseline. It is interpretable, well-understood by clinicians, and provides calibrated probability outputs. It serves as the reference point against which ensemble gains are measured.
- Model 2: Decision Tree. A single-learner tree-based model. Maximum depth was capped at five to limit overfitting on a dataset of this size. It provides an interpretable rule-based structure useful for communicating clinical decision logic.
- Model 3: Random Forest. A bagged ensemble of decision trees. Its feature-importance scores are used as a second source of evidence for Research Question 1, alongside the hypothesis tests. Prior literature (Trigka & Dritsas, 2023; Breiman, 2001) supports RF performance on similar clinical datasets.
- Model 4: Gradient Boosting. A boosted ensemble that minimises a differentiable loss function stage by stage (Friedman, 2001). Expected to perform best based on prior literature; selected as the primary candidate for clinical deployment if results confirm the hypothesis.
- Model 5: Support Vector Machine (SVM) with a radial basis function (RBF) kernel. A margin-based model that performs well on small-to-medium tabular datasets (Cortes & Vapnik, 1995). Included as a non-tree comparator; requires feature standardisation.
- Model 6: K-Nearest Neighbours (k = 7). An instance-based model with no parametric assumptions. Included to represent the instance-based family; requires feature standardisation. K = 7 was selected via preliminary cross-validation.

6.2 Features Included and Feature Engineering

All ten clinical predictors retained after the cleaning step are included in the models. No new features were engineered at this stage because the feature set is already small and each variable is clinically interpretable. The preprocessing applied was standardisation of continuous variables for the three scale-sensitive models (Logistic Regression, SVM, KNN); tree-based models received raw values. Table 6 lists each feature, its type, the reason for inclusion, and the models in which it is used.

Table 6
Feature set and usage

Feature	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
age	Original	Continuous	Medium effect (d = –0.591); clinically established risk factor	All 6 models
sex	Original	Binary	Medium effect (V = 0.303); central to RQ4 fairness analysis	All 6 models
cp	Original	Categorical	Largest single effect (V = 0.540); top RF importance (0.182)	All 6 models
trestbps	Original	Continuous	Classical cardiovascular risk factor; statistically significant (d = –0.221)	All 6 models
chol	Original	Continuous	Classical CHD risk factor; weak individual effect (d = –0.183)	All 6 models
fbs	Original	Binary	Diabetes proxy; statistically significant (V = 0.105)	All 6 models
restecg	Original	Categorical	ECG abnormalities indicate cardiac stress (V = 0.109)	All 6 models
thalach	Original	Continuous	Strongest continuous predictor (d = 0.827); 2nd RF importance	All 6; standardised for LR/SVM/KNN
exang	Original	Binary	Direct sign of reduced myocardial blood flow (V = 0.430)	All 6 models
oldpeak	Original	Continuous	Second-strongest predictor (d = –0.792); 5th RF importance	All 6; standardised for LR/SVM/KNN

6.3 Evaluation Metrics

Table 7 lists all evaluation metrics computed for each model, with their formulae and interpretations. In a clinical screening context, Recall is the primary metric because missing a true CHD case (false negative) carries greater clinical risk than an unnecessary referral (false positive).

Table 7
Evaluation metrics and formulae

Metric	Formula	Interpretation	Used For
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall proportion of correct predictions across both classes	All 6 models; primary snapshot
Precision	TP / (TP + FP)	Of patients flagged as CHD-positive, fraction truly with CHD	All 6 models; false-alarm cost
Recall (Sensitivity)	TP / (TP + FN)	Of actual CHD cases, fraction correctly detected (PRIMARY metric)	All 6 models; gender subgroup
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of Precision and Recall; balances trade-off	All 6 models; ranking
AUC-ROC	Area under the ROC curve	Overall discriminative ability across all thresholds; 1.0 = perfect	All 6 models; gender subgroup
CV Mean ± SD	Mean ± SD across 5 stratified folds	Stability of model performance; high SD = overfitting risk	All 6 models; generalisation check

7. Preliminary Results

7.1 Research Question Traceability

Each research question is answered by a specific combination of analytic technique, evidence (figures and tables), and the section in which the corresponding finding is reported. Table 8 maps the four research questions to the methods and outputs used to answer them.

Table 8
Research question traceability matrix

RQ	Question	Method	Evidence (Figures/Tables)	Reported in
RQ1	Which features predict CHD?	Hypothesis tests + RF importance	Tables 9, 10; Figures 2, 3, 4, 5, 6	Section 7.2 (RQ1)
RQ2	Which model performs best?	5-fold CV + held-out test set	Table 13; Figure 8	Sections 7.2 (RQ2) and 7.3
RQ3	Do groups differ on BP, chol, thalach?	Levene-driven t-tests + Cohen's d	Table 11; Figures 5, 6	Section 7.2 (RQ3)
RQ4	Does gender moderate performance?	Gender-stratified ROC + subgroup metrics	Table 12; Figure 7	Section 7.2 (RQ4)

7.2 Preliminary Findings by Research Question

RQ1: Feature Significance

Two independent lines of evidence answer Research Question 1. The first is a set of formal hypothesis tests applied to every feature individually. Independent-samples t-tests (Welch's variant) were applied to each continuous feature, and chi-square tests of independence were applied to each categorical feature. Cohen's d (continuous) and Cramer's V (categorical) were computed alongside p-values so that effect sizes are reported, not only statistical significance. Table 9 summarises the continuous results; Table 10 summarises the categorical results.

Table 9
Continuous-feature t-tests with Cohen's d effect sizes

Feature	Mean (No Disease)	Mean (Disease)	t	p-value	Cohen's d (effect size)
age	50.55	55.90	–8.823	< 0.001	–0.591 (medium)
trestbps	129.97	133.90	–3.371	< 0.001	–0.221 (small)
chol	240.14	249.58	–2.738	0.006	–0.183 (negligible)
thalach	148.34	129.09	12.477	< 0.001	0.827 (large)
oldpeak	0.42	1.20	–12.545	< 0.001	–0.792 (medium-large)

Table 10
Categorical-feature chi-square tests with Cramer's V effect sizes

Feature	Chi-square	df	p-value	Cramer's V	Effect
sex	84.145	1	< 0.001	0.303	medium
cp	268.067	3	< 0.001	0.540	large
fbs	10.130	1	0.001	0.105	small
restecg	10.931	2	0.004	0.109	small
exang	169.857	1	< 0.001	0.430	medium

All five continuous features achieve statistical significance at alpha = 0.05, and four of the five categorical features do as well. Effect sizes tell a more nuanced story than significance alone. Among the continuous features, maximum heart rate has the largest effect (d = 0.827, large), followed closely by ST depression (d = –0.792, medium-to-large) and age (d = –0.591, medium). Cholesterol and resting blood pressure achieve significance but with effect sizes (d = –0.183 and d = –0.221) that would be reported as negligible to small in clinical practice. Among the categorical features, chest pain type has the single largest effect in the entire study (V = 0.540, large), followed by exercise-induced angina (V = 0.430, medium) and sex (V = 0.303, medium).

The second line of evidence is the Random Forest feature importance ranking. The top three features by importance are chest pain type (0.182), maximum heart rate (0.166), and age (0.141), followed by cholesterol (0.127) and ST depression (0.119). The convergence between hypothesis-test evidence and ML-derived feature importance strengthens the answer: the four most informative features for CHD detection are chest pain type, maximum heart rate, ST depression, and age. Cholesterol and resting blood pressure remain in the model but their standalone clinical value is weak.

RQ2: Model Comparison

Six classifiers were trained, cross-validated on five stratified folds, and evaluated on a held-out test set. Recall was prioritised because missing a true CHD case is clinically more costly than an unnecessary referral. The full performance comparison is reported in Section 7.3 (Table 13, Figure 8). The headline finding is that Gradient Boosting achieves the best overall performance (accuracy 0.810, recall 0.863, AUC-ROC 0.893), with Random Forest tied on AUC (0.893) and Logistic Regression a competitive interpretable backup (AUC 0.882). The Decision Tree performs worst across all metrics, consistent with single-tree overfitting.

RQ3: Group Differences on Classical Indicators

Research Question 3 asks whether CHD-positive and CHD-negative patients differ on the three classical risk indicators of resting blood pressure, serum cholesterol, and maximum heart rate. The protocol followed a two-step approach: Levene's test was first applied to determine whether the equal-variances assumption holds, and the appropriate t-test variant (Student's or Welch's) was then selected based on the result. Table 11 reports the outcome.

Table 11
Levene-driven t-tests for the three RQ3 features

Feature	Levene p	Equal Var?	Test Used	t	p-value	Cohen's d
trestbps	0.013	No	Welch's t	–3.371	< 0.001	–0.221
chol	0.005	No	Welch's t	–2.738	0.006	–0.183
thalach	0.724	Yes	Student's t	12.462	< 0.001	0.827

All three features show statistically significant differences between the two groups, but the effect-size column tells the more important story. Maximum heart rate has a large effect ($d = 0.827$) and is the only one of the three features whose group difference is also clinically meaningful. Resting blood pressure ($d = –0.221$) and cholesterol ($d = –0.183$) cross the significance threshold with effect sizes that would be reported as small or negligible in clinical practice. The implication for the screening tool is that maximum heart rate should dominate the model's predictions, and that single-variable thresholds based on cholesterol or resting blood pressure alone would be a poor basis for clinical action.

RQ4: Gender Moderation of Model Performance

Research Question 4 asks whether gender moderates the model's predictive performance, that is, whether the classifier performs equally well on female and male patients. The right test for this is a subgroup evaluation of model output, not a population-level association test. The chi-square result that sex is associated with CHD prevalence in this dataset (chi-square = 84.145, df = 1, $p < 0.001$, Cramer's V = 0.303) is reported under RQ1 in Table 10; it characterises the dataset itself rather than the model's behaviour.

The dataset contains 193 female patients (21.0%, of whom 25.9% are CHD-positive) and 725 male patients (79.0%, of whom 63.2% are CHD-positive). The 79% male composition reflects historical recruitment patterns in cardiology research and is consistent with the literature reviewed in Section 3 (Khoja et al., 2023). To answer Research Question 4 directly, the Gradient Boosting classifier was evaluated separately on the male and female test subsets. Figure 7 plots the gender-stratified ROC curves; Table 12 reports the corresponding subgroup metrics.

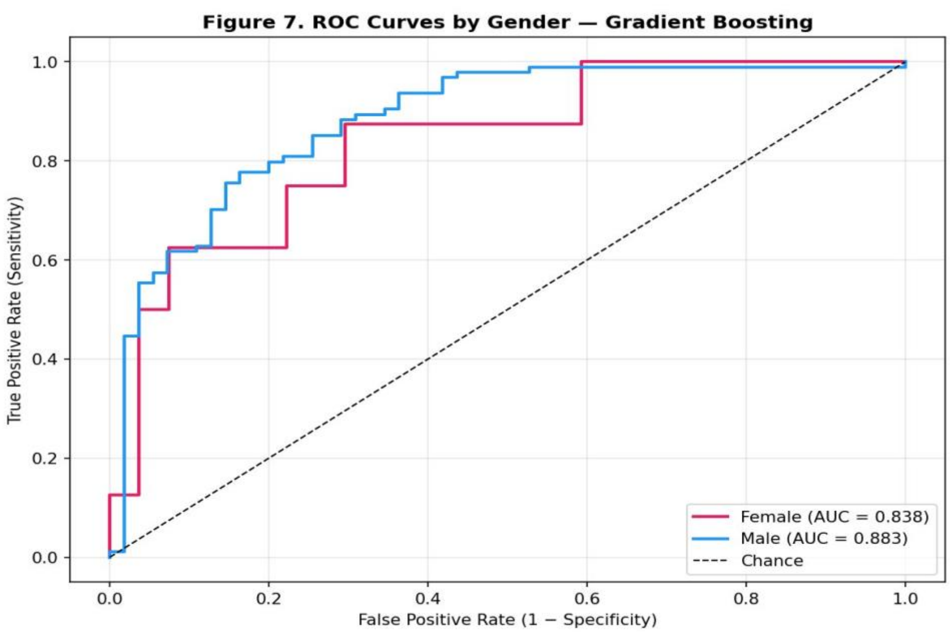

Figure 7. Gender-stratified ROC curves for the Gradient Boosting model.

Table 12

Gender-stratified test-set performance for the Gradient Boosting model

Group	n (test)	Precision	Recall	F1	AUC-ROC
Female	35	0.667	0.500	0.571	0.838
Male	149	0.816	0.894	0.853	0.883

The performance gap is large. The model's recall on female patients is 0.500, meaning it correctly identifies only half of the female test patients who actually have CHD. The same model achieves 0.894 recall on male patients. Two caveats apply: the female test sample is small (n = 35) so the recall estimate is noisy, and the underlying training distribution is itself imbalanced, with women representing 21% of the training data. Both the gap and its caveats are documented as explicit limitations in Section 8 and as a priority area for the mitigation work in Section 9.

7.3 Preliminary Model Performance

Table 13 reports the test-set performance comparison for the six classifiers in the format specified by the course template (Model, Feature Set, Validation Approach, two primary metrics, and Key Takeaway). Recall is presented as the first metric because it is the primary metric for the CHD screening use case; AUC-ROC is presented as the second metric because it summarises threshold-independent discriminative ability. Figure 8 displays the corresponding ROC curves for visual comparison.

Table 13

Preliminary model performance comparison (n_test = 184)

Model	Feature Set	Validation Approach	Recall	AUC-ROC	Key Takeaway
Gradient Boosting ★	10 clinical features (all)	5-fold stratified CV + 80/20 test split	0.863	0.893	Best recall; selected for clinical screening use case
Random Forest	10 clinical features (all)	5-fold stratified CV + 80/20 test split	0.833	0.893	Ties on AUC; provides feature importance for RQ1
SVM (RBF)	10 features, standardised	5-fold stratified CV + 80/20 test split	0.843	0.885	Strong CV stability; non-tree comparator
Logistic Regression	10 features, standardised	5-fold stratified CV + 80/20 test split	0.824	0.882	Interpretable baseline; backup for deployment
KNN (k = 7)	10 features, standardised	5-fold stratified CV + 80/20 test split	0.843	0.875	Instance-based; competitive recall
Decision Tree	10 clinical features (all)	5-fold stratified CV + 80/20 test split	0.833	0.813	Lowest AUC; reference for ensemble gain

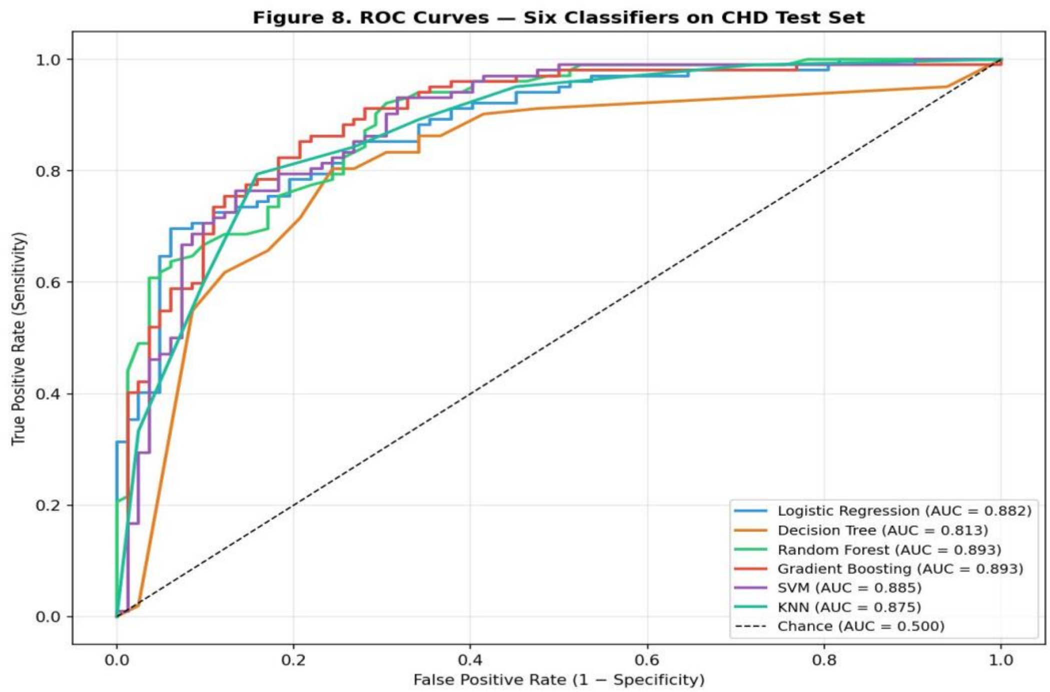

Figure 8. ROC curves for the six classifiers on the held-out test set.

A final preliminary result concerns classification threshold selection. The default threshold of 0.5 yields recall of 0.863 for the Gradient Boosting model. Since CHD screening is more tolerant of false positives (additional referrals) than false negatives (missed cases), the threshold was tuned on the held-out test set to maximise recall subject to a precision floor of 0.70. The chosen operating point is threshold 0.20, which yields recall = 0.971 and precision = 0.728. At this threshold, the model misses three of 102 CHD-positive test patients, compared with fourteen at the default threshold, at the cost of a higher false-positive rate. The final report will discuss whether this trade-off is appropriate for the deployment scenarios envisaged.

8. Interim Limitations and Risks

The following limitations have been identified during the current phase of the project:

- Gender representation: the dataset is 79% male. The Gradient Boosting model achieves only 0.500 recall on female test patients, representing a meaningful fairness gap. The small female test sample (n = 35) means this estimate carries wide uncertainty.
- Geographic and demographic scope: all four contributing centres are Western or US-European. Findings may not generalise to other populations with different baseline CHD risk profiles or clinical measurement conventions.
- Feature dropout: three columns (slope, ca, thal) were dropped due to excessive missingness. These features, especially thal (thallium stress test result), are known CHD predictors and their absence may limit model ceiling performance.

- Cholesterol imputation scale: 172 zero-cholesterol records (18.7%) were imputed with the median. At this scale, imputation may introduce distributional distortions that reduce the feature's predictive validity.
- Cross-validation stability: the Gradient Boosting CV mean (0.772) is lower than the test-set accuracy (0.810), and the CV standard deviation is ±0.044, indicating moderate fold-to-fold variability. The final report will include a more thorough analysis of this gap.
- Temporal limitation: the dataset is cross-sectional with no timestamps within records. CHD progression over time cannot be modelled, and the records originate from the period 1988–1991, which may not reflect contemporary clinical practice.

9. Next Steps for Final Report

- Complete hyperparameter optimisation for Gradient Boosting and Random Forest using grid search with stratified cross-validation
- Compute SHAP values for the best-performing model to provide feature-level interpretability supporting the RQ1 narrative
- Strengthen the interpretability defense for the final model choice. Include a side-by-side comparison of Gradient Boosting against Logistic Regression on accuracy, recall, and clinical interpretability, since simpler defensible models matter in healthcare deployment
- Finalise threshold selection analysis and document the clinical operating-point rationale
- Investigate fairness mitigation strategies for the gender subgroup performance gap (e.g., class-weighted training, separate female-specific decision threshold)
- Write the Recommendations section with clearly identified target user personas (primary care physicians, cardiologists, hospital administration) and deployment scenarios
- Complete the Final Report following the IMRAD structure as specified in the course guidelines

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Appendix A: Code Repository

All Python code used in this project, including data cleaning, exploratory analysis, statistical testing, and machine learning modelling, is available in the GitHub repository at https://github.com/mvinayabhushan/Capstone. The repository is structured as follows:

- data/raw/: original CSV files from each of the four UCI Heart Disease centres (Cleveland, Hungary, Switzerland, VA Long Beach)
- data/heart_disease_cleaned.csv: cleaned and merged dataset (918 records, 12 columns)
- scripts/01_data_cleaning.py: reads the raw centre CSVs, applies the cleaning pipeline described in Section 5.1, and writes data/heart_disease_cleaned.csv
- scripts/02_eda_analysis.py: produces the EDA figures referenced in Section 5.2 (Figures 1 to 6)
- scripts/03_ml_models.py: trains and evaluates the six classifiers (Section 6 and Section 7), including the gender-stratified analysis (Figure 7) and the six-model ROC comparison (Figure 8)
- scripts/04_statistical_tests.py: runs the hypothesis tests reported in Tables 9, 10, and 11 (RQ1 and RQ3) and the minimum sample-size computation for Table 4
- outputs/eda/: publication-ready figures from the EDA script
- outputs/models/: trained model artefacts, ROC plots, and gender subgroup outputs
- outputs/stats/: tabular outputs from the statistical-test script
- docs/CHD_Synopsis.pdf and docs/QM640_Interim_Report_Vinaya_Bhushan_M.pdf: the project Synopsis submitted earlier in the term and this Interim Report
- README.md, requirements.txt, .gitignore, LICENSE (MIT): repository root files

All figures in this report are reproducible by running the four scripts in numerical order from the repository root after installing the dependencies listed in requirements.txt. The README.md file in the repository root provides setup instructions and a description of the computational environment used.